Project Proposal

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Chosen topic: The project will investigate the design and performance of a converging-diverging rocket nozzle for a NASA-style propulsion system, with emphasis on how nozzle geometry affects thrust and optimal expansion over a range of altitudes. A converging-diverging nozzle accelerates high-pressure combustion gases to supersonic speeds, converting thermal energy into directed momentum that produces thrust. However, nozzle performance depends strongly on ambient pressure: a nozzle designed for sea level becomes underexpanded at high altitude, while one designed for vacuum becomes overexpanded at low altitude, reducing performance and potentially introducing shocks.

The assignment will begin with isentropic nozzle relations to compute exit properties and thrust as functions of area ratio, and will then incorporate an atmospheric pressure model to evaluate altitude-dependent performance and the resulting change in nozzle design. This project connects quasi-one-dimensional compressible flow relations with a practical propulsion design problem, allowing students to: (1) determine exit conditions as functions of area ratio, (2) identify design area ratios for different operating conditions, (3) evaluate performance when the nozzle operates away from its design altitude, and (4) determine conditions where overexpanded operation leads to shock formation within the nozzle.

Assignment skeleton:

1. Define reservoir conditions and compute supersonic exit Mach number as a function of nozzle exit area ratio using the isentropic area-Mach relation.
2. Compute exit pressure, temperature, and velocity for each area ratio and plot the results.
3. Incorporate an atmospheric pressure model and use the rocket thrust equation to calculate thrust as a function of area ratio across a range of altitudes.
4. Determine the design condition where exit pressure equals ambient pressure and identify the corresponding area ratio for each altitude.
5. For selected nozzle area ratios, evaluate thrust over the same altitude range and compare performance when the nozzle operates away from its design altitude.
6. Identify conditions where overexpanded operation causes the isentropic solution to break down, determine when a normal shock would form in the nozzle, and compare thrust with and without shock formation.

Resources:

- Anderson, *Modern Compressible Flow: With Historical Perspective*, 4th edition, 2021 McGraw Hill:
    - Chapter 5: Quasi-One-Dimensional Flow
    - Loosely based on Example 5.3: Extended to examine thrust variation and optimal expansion at different altitudes.
- NASA Glenn Research Center: Thrust Equations Summary (https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/thrust-equations-summary/)
- NASA Glenn Research Center: Earth Atmosphere Equation – Metric (https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/earth-atmosphere-equation-metric/)
- Equations from Module 2a (general propulsion device thrust), 3a (area-Mach relation and stagnation properties), and 4b (CD nozzle with normal shocks) notes